

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 2) - Single Correct

Find out the surface charge density at the intersection of point $x = 3$ m plane and x -axis, in the region of uniform line charge of 8 nC/m lying along the z -axis in free space.

(1) 0.424 nCm^{-2}

(2) 47.88 C/m

(3) 0.07 nCm^{-2}

(4) 4.0 nCm^{-2}

Q2: 17 March (Shift 2) - Numerical

The electric field in a region is given by

$$\vec{E} = \frac{2}{5}E_0\hat{i} + \frac{3}{5}E_0\hat{j} \text{ with } E_0 = 4.0 \times 10^3 \frac{\text{N}}{\text{C}}.$$

flux of this field through a rectangular surface

area 0.4 m^2 parallel to the $Y - Z$ plane is _____ Nm^2C^{-1}

Q3: 18 March (Shift 1) - Single Correct

A plane electromagnetic wave of frequency 100MHz is travelling in vacuum along the x - direction. At a

particular point in space and time, $\vec{B} = 2.0 \times 10^{-8}\hat{k}\text{T}$. (where, $\hat{k}$ is unit vector along z -direction) What is $\vec{E}$ at this point?

(1) $0.6\hat{j} \text{ V/m}$

(2) $6.0\hat{k} \text{ V/m}$

(3) $6.0\hat{j} \text{ V/m}$

(4) $0.6\hat{k} \text{ V/m}$

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Answer Key

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Q1 (1)

Q2 (640)

Q3 (3)

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Hints and Solutions

MathonGo

Q1 (1)

$$\frac{2K\lambda}{r} = \frac{\sigma}{\epsilon_0} \quad (x = 3 \text{ m})$$

$$\sigma = 0.424 \times 10^{-9} \frac{\text{C}}{\text{m}^2}$$

Q2 (640)

$$\phi = E_x A \Rightarrow \frac{2}{5} \times 4 \times 10^3 \times 0.4 = 640$$

Q3 (3)

$$E = BC = 6$$

$$(\text{Dir. of wave}) \parallel (\vec{E} \times \vec{B})$$

$$\hat{i} = \hat{j} \times \hat{k}$$

$$\vec{E} = 6\hat{j} \text{ V/m}$$